

CIS 588 - Neural Computing, Fall 2026

Mon, Wed, and Fri, 1 – 1:50 pm, DION 311

Instructor: Dr. Debarun Das

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Office hours: Mon 1:50 to 3:50 pm; Wed 1:50 to 2:50 pm; Fri 1:50 to 2:50 pm

TA/Grader: TBD

1. Class Time and Classroom Location

The lectures will be held in DION 311 on Mondays, Wednesdays, and Fridays from 1 to 1:50 pm.

2. Required Textbook

There is no required textbook for this course. The required materials (slides, codes, notes, etc.) will be provided by the instructor.

3. Textbooks for Reference

The following books can be used as a reference for this course.

- Harrison Kinsley and Daniel Kukiela. *Neural Networks from Scratch in Python: Building Neural Networks in Raw Python*. Publication Year: 2020.
- Charu C. Aggarwal, *Neural networks and deep learning*. Vol. 10. No. 978. Publisher: Springer, 2018.
- Yoshua Bengio, Ian Goodfellow, and Aaron Courville. *Deep learning*. Vol. 1. Cambridge, MA, USA: MIT press, 2017. (<https://www.deeplearningbook.org/>)

4. Prerequisites

Any course in programming languages and data structures.

5. Course Objectives

- To understand the core concepts and mathematical foundations of artificial neural networks
- To understand and implement artificial neural networks from scratch.
- To understand, compare, and analyze different optimization strategies of neural networks.
- To explore the structure and applications of advanced neural network architectures

6. Course Outcomes

(Students should be able to)

- Build and train multi-layer neural networks from scratch using NumPy in Python.
- Implement forward and backward propagation from scratch, using the chain rule and gradients.
- Apply and compare optimization techniques such as ADAM, RMSProp, and Momentum.
- Describe the structure and applications of the most common deep learning architectures.

7. Catalog Description

Fundamentals of artificial neural networks including application needs for neural networks, discussing the various architectures, learning algorithms and examples of applications. The standard neural networks are discussed in greater details, which allows for branching of architectures and combining of strategies for growing and/or constructing neural networks.

8. Important Dates and Grading

The *tentative* examination schedule is as follows:

- Midterm exam: Monday, October 26, 2026 (Tentative and subject to change, in-person)
- Final exam*: Tuesday, December 15, 2026, 3 pm to 6 pm (in-person)

*: See Final exam schedule at <https://www.umassd.edu/registrar/final-exams/>

Your numerical scores will be based upon the following contributions:

- 7% Class attendance and participation
- 25% Quizzes
- 33% Homework Assignments
- 12% Midterm exam
- 23% Final exam (Comprehensive)

The letter grades will be assigned using the following *approximate* scale. These are tentative and are subject to change, if required:

Letter Grades	Percentage
A-, A, A+	90 to 100
B-, B, B+	80 to 89.99
C-, C, C+	70 to 79.99
D-, D, D+	60 to 69.99
F	0 to 59.99

Academic dishonesty will be “rewarded” with a grade of “F”. **Sharing of solutions to homework problems is strictly prohibited.**

Students may not elect the P/NC option for a course in Fall 2026. All courses will be evaluated using the UMass Dartmouth grading system as defined here:

https://catalog.umassd.edu/content.php?catoid=62&navoid=5015#Grades_and_Grading_System

9. Course Requirements

Class attendance is *mandatory*. You are responsible for all announcements and all material presented in the lecture sessions.

In addition, students are responsible for all official correspondence sent to their standard UMD e-mail address (@umassd.edu).

This course will have assignments posted on Canvas. Each assignment contains a number of questions that may require written answers, and possibly writing programs using the Python programming language. Each

assignment must be submitted online through the [Canvas](#) learning management system, unless stated otherwise.

Each assignment will have a deadline (date and time). Late submissions of quizzes and assignments will *not* be considered unless *explicitly* specified in class (or approved by the course instructor). **It is your responsibility to make sure that you submit the right files in a timely manner.**

The course will have online quizzes posted on Canvas. The due dates of these online quizzes will be announced as soon as they are posted. There will also be a few short, in-person quizzes spread throughout the semester. Dates for such quizzes will be announced beforehand.

The midterm and final exams are in-person and on paper. There will be no make-up midterm and final exam. If you have a conflict with the midterm or final exam, contact the instructor well in advance so that your situation can be evaluated, and a solution can be found. Of course, only serious conflicts will be considered.

Any requests for re-grading your homework assignments or exams must be submitted in *writing*, along with the assignment or exam, within one week from the day it was returned. Your request must detail what you think was incorrect about the grading and must be submitted to TA/grader by email.

10. Academic Honesty

You must individually design and write your own solutions for all assignments unless group work is explicitly permitted. While you may consult external websites to deepen your understanding, **you are NOT permitted to copy and paste code, text, or solutions** from any online source, including forums (e.g., Stack Overflow), code-sharing platforms (e.g., GitHub), AI-generated content, or solution databases (e.g., Chegg, CourseHero).

If you consult any external resources, you must clearly cite them by listing all URLs or sources you referred to. Failure to do so will be considered a violation of academic integrity.

Students should be aware that suspect assignments (e.g., those without drafts, without works cited pages, or with large departures in style) will be submitted to [SafeAssign](#) by the instructor for the purpose of detecting possible plagiarism. Submitted assignments will be included in the UMass Dartmouth dedicated databases of assignments. These databases of assignments will be used solely for the purpose of detecting possible plagiarism during the grading process during this term and in the future. Students must provide an electronic copy of their assignment to the instructor for submission to the service when plagiarism is suspected, to receive a grade on the assignment and to avoid possible sanctions.

Using someone else's work, code, or ideas—whether from a peer or an online source—without proper citation constitutes *plagiarism*. This includes paraphrasing code or solutions without credit. The penalties for plagiarism range from receiving an “F” on the assignment or in the course to possible expulsion from the university. For further information about the University's policy on Academic Dishonesty and about issues of Academic Integrity, please refer to the following link:

<https://www.umassd.edu/policies/active-policy-list/academic-affairs/academic-integrity-policy-and-reporting-form/>

11. Center for Access and Success

In accordance with University policy, if you have a documented disability and require accommodations to obtain equal access in this course, please meet with the instructor at the beginning of the semester and provide the appropriate paperwork from the [Center for Access and Success](#). The necessary paperwork is obtained when you bring proper documentation to the Center.

12. Title IX Information

The purpose of a university is to disseminate information, as well as to explore a universe of ideas, to encourage diverse perspectives and robust expression, and to foster the development of critical and analytical thinking skills. In many classes, including this one, students and faculty examine and analyze challenging and controversial topics. If a topic covered in this class triggers post-traumatic stress or other emotional distress, please discuss the matter with the professor or seek out confidential resources available from the Counseling Center, 508-999- 8648 or - 8650, or the Victim Advocate in the Center for Women, Gender and Sexuality, 508-910-4584. In an emergency, contact the Department of Public Safety at 508-999-9191, 24 hrs./day.

UMass Dartmouth, following national guidance from the Office of Civil Rights, requires that faculty follow UMass Dartmouth policy as a "mandated reporter" of any disclosure of sexual harassment, abuse, and/or violence shared with the faculty member in person and/or via email. These disclosures include but are not limited to reports of sexual assault, relational abuse, relational/domestic violence, and stalking.

While faculty are often able to help students locate appropriate channels of assistance on campus, disclosure by the student to the faculty member requires that the faculty member inform the University's Title IX Coordinator in the Office of Diversity, Equity and Inclusion at 508-999-8008 to help ensure that the student's safety and welfare are being addressed, even if the student requests that the disclosure not be shared.

For confidential counseling support and assistance, please go to <http://www.umassd.edu/sexualviolence/>

13. Academic and Technical Support

Tutoring: If you have difficulty with the coursework, please reach out to me, your TA, or contact the [Academic Resource Center](#).

Technical Help: 24/7 email, live chat, and phone support for *myCourses* is available at the [Canvas support portal](#). Do you need help with other UMass Dartmouth technologies? [Please contact CITS](#).

Web-conference with Zoom: <https://instructionaldev.umassd.edu/zoom/>

14. Tentative Schedule

The following is a tentative schedule of the topics to be covered in this course. Please note that the schedule and the topics below are subject to change.

Dates	Topics
Sept. 2 – 4	Introduction to Neural Networks, Neurons and Layers. Python Overview.
Sept. 7 – 11	Linear Algebra Recap. Dot product. Multi-Layer Neural Networks Implementation. <i>Labor Day on Monday, September 07. No class that day.</i>
Sept. 14 – 18	Neural Network activation functions
Sept. 21 – 25	Neural Network Forward Pass. Cross Entropy Loss.
Sept. 28 – Oct 2	Optimization in Neural Networks
Oct. 5 – 9	Derivatives, Gradient, and Partial Derivatives
Oct. 12 – 16	Chain Rule. Backpropagation on a Single Neuron <i>Indigenous Peoples' Day on Monday, October 12. No class that day.</i> <i>Class on Tuesday, October 13</i>
Oct. 19 – 23	Backpropagation on multi-layer NNs. Midterm Exam Review
Oct 26 – 30	Midterm Exam (Tentative). Role of Matrices in Backpropagation. <i>Mental Health Day on Wednesday, October 28. No class that day.</i>
Nov. 2 – 6	Backpropagation on RELU activation class and cross entropy loss.
Nov. 9 – 13	Backpropagation on SoftMax Activation. Building the backpropagation pipeline.
Nov. 16 – 20	Learning Rate Decay, Momentum, ADAGRAD optimizer. ADAM Optimizer, RMSProp Optimizer.
Nov. 23 – 27	L1/L2 Regularization. Dropout Layers. <i>Thanksgiving Break.</i>
Nov. 30 – Dec. 4	Deep Learning Architectures and Applications.
Dec. 7 – 9	Final Exam Review
Dec. 15	Final Exam (Comprehensive, in-class) 3 pm to 6 pm

15. Important Dates

Some of the important dates are:

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|---------------------------------|-------------------------------|
| • Wednesday, September 14, 2026 | Add, Drop, and Audit deadline |
| • Wednesday, October 8, 2026 | Pass/Fail Deadline |
| • Friday, November 13, 2026 | Course Withdrawal Deadline |

The complete academic calendar is on the UMassD website:

<http://www.umassd.edu/academiccalendar/>

16. Syllabus Disclaimer

This syllabus is a guide, and every attempt is made to provide an accurate overview of the course. However, circumstances and events may make it necessary for the instructor to modify the syllabus during the semester and may depend, in part, on the progress, needs, and experiences of the students. Changes to the syllabus will be made with notice.